+91 9043916443
Chennai, India
melvinimmanuelg@gmail.com

Melvin Immanuel Gregory

Portfolio Link
github.com/Melvin0163
AI Engineer | ML Engineer | Data Scientist linkedin.com/in/melvinimmanuelg

PROFESSIONAL SUMMARY

Data Science postgraduate with a strong mathematical foundation and hands-on experience in Artificial Intelligence, Machine Learning, Deep Learning, NLP, and Large Language Models. Experienced in building Agentic RAG systems, developing deep learning architectures, evaluating AI-generated outputs, and optimizing retrieval pipelines. Focused on developing reliable and scalable AI solutions through mathematical reasoning, experimentation, and data-driven engineering.

TECHNICAL SKILLS

Programming	Python, SQL, R, Java, C++, PL/SQL
Machine Learning	Scikit-learn, Regression, Classification, Clustering, Model Evaluation
Deep Learning	TensorFlow, PyTorch, Keras, GANs, BiLSTM, Neural Networks
NLP & LLMs	NLP, RAG, LangChain, LangGraph, Prompt Engineering, Embeddings
Document AI	EasyOCR, OCR, PDF Processing, Document Extraction, Semantic Search
Data Libraries	Pandas, NumPy, SciPy, Matplotlib, Seaborn
Data Visualization	Power BI, Tableau, Advanced Excel, VBA, Power Query
Tools & Frameworks	Git/GitHub, FastAPI, Flask, Jupyter Notebook
Cloud & Platforms	AWS (Basics), Google Colab, Hadoop
Core Concepts	Time Series Forecasting, Statistical Modeling, Information Retrieval

PROJECTS AND INTERNSHIPS

Handshake AI – AI Analyst *Jul 2026 – Present*

- Evaluated AI-generated outputs for factual accuracy, logical reasoning, relevance, and instruction adherence.
- Contributed to mulitiple STEM projects, assessing mathematical reasoning and problem-solving capabilities.
- Performed structured error analysis and model evaluation to identify weaknesses and improve AI response reliability.

Legal Document Intelligence Using Agentic AI GitHub

- Architected and benchmarked two legal AI pipelines: Agentic RAG (BM25 + dense retrieval) and JEV-compatible typed-decision reasoning with Open Knowledge Format (OKF).
- Engineered multi-stage agentic workflows for document parsing, clause identification, query routing, statutory navigation, evidence retrieval, and citation validation using LangGraph and LLMs.
- Achieved 81% lower LLM token consumption, improved routing accuracy from 72% to 94%, and increased expected-section retrieval from 83% to 92%.

Adversarial Time Series Forecasting Using GANs GitHub

- Developed GAN, Shared GAN, and WGAN-GP architectures integrated with BiLSTM for time-series imputation and forecasting.
- Processed 76 years of climate data using preprocessing, normalization, missing-value simulation, and chronological validation.
- Achieved an R^2 score of 0.85 for 7-day temperature forecasting using quantitative model evaluation.

Schneider Electric – Data Analytics Intern *Dec 2024 – Jan 2025*

- Developed interactive Power BI dashboards to visualize business KPIs and support data-driven decisions.
- Automated Excel reporting workflows using VBA macros to reduce repetitive manual reporting tasks.
- Performed data cleaning, transformation, and analysis using Excel and Power Query.

EDUCATION

M.Sc. Data Science *Jun 2025 – Jun 2027*
Vellore Institute of Technology (VIT), Chennai
CGPA: 9.02/10
Relevant Coursework: Machine Learning, Deep Learning, Statistical Modeling, Data Mining, Exploratory Data Analysis, Big Data Analytics, Cloud Computing, Business Analytics

B.Sc. Mathematics *Jun 2022 – Jun 2025*
Loyola College, Chennai
CGPA: 9.22/10 (Top 5%)
Relevant Coursework: Linear Algebra, Probability Theory, Statistical Inference, Regression Analysis, Numerical Methods, Design and Analysis of Algorithms

CERTIFICATIONS & ACHIEVEMENTS

- Stanford Online – Supervised Machine Learning: Regression and Classification
- Stanford Online – Unsupervised Learning, Recommenders and Reinforcement Learning
- Introduction to MongoDB – Simplilearn, Data Structures in C++ – Scaler Topics